

ADIL ARYA

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EDUCATION

Columbia University <i>Master of Science in Artificial Intelligence</i>	New York City, New York Sep 2026 – Dec 2027
University of Minnesota - Twin Cities <i>Bachelor of Science in Computer Science</i> • Honors: 3.912 / 4.0 GPA, 5x CSE Dean's List, High Honors • Courses: Algorithms & Data Structures, Operating Systems, Machine Learning, Computer Vision, Robotics, NLP, Artificial Intelligence I & II • B.S. completed in 2 years (90+ credits, 8 graduate-level courses).	Minneapolis, Minnesota Sep 2024 – May 2026

WORK EXPERIENCE

Minnesota Robotics Institute <i>Undergraduate Researcher</i> • Built a 3D-reconstruction evaluation pipeline benchmarking GLOMAP vs COLMAP across 30 trials (Chamfer, precision/recall, F-score); raised F-score 0.79 to 0.89 and cut Chamfer error ~30%, driving the lab's adoption of GLOMAP as its reconstruction backbone (Python, NumPy, Pandas). • Architected CableMind, an autonomous robotic system pairing a modular state machine (perception to planning to execution via xArm SDK) with a LoRA-fine-tuned OpenVLA (Llama-2-7B) policy mapping language instructions to robot trajectories; 5/5 success on single-clip, ~60% on randomized 4-clip routing.	Minneapolis, Minnesota Oct 2024 – May 2026
Divertica Inc. <i>Computer Vision Intern</i> • Shipped an end-to-end ML pipeline processing 4K/60fps video into structured retail planograms: frame stitching, object detection, and OCR stages wired into one system. • Fine-tuned and evaluated detection models (YOLOv8/v12 on SKU110K, ResNet50, OCR) to 0.87 F1 / 0.90 mAP50; found v12 gave no real gain over v8, then diagnosed a ~630K-class scaling bottleneck, and redirected the team towards a retrieval-based design.	Remote Jun 2025 – Aug 2025

PROJECTS

TrialFind <i>Origin House O1 Summit Hackathon</i> • 1st Place, TinyFish Sponsor Track (24-hour build, team of 3): agentic system matching cancer patients to clinical trials, pairing an LLM summarization agent with parallel browser agents searching live trial registries. • Orchestrated 6 concurrent browser agents (up from 2) across 3 API providers, parallelizing real-time multi-website search across trial registries.	Apr 2026 – Apr 2026
Physical IDE <i>Google I/O Build with AI Hackathon x Google Cloud Labs</i> • Finalist, top 7 of 50+ teams (12-hour build, team of 3): real-time hardware assembly verification agent streaming webcam frames to Gemini Live API, collapsing 3 model calls (vision, reasoning, speech) into one round-trip. • Owned the serving backend: a FastAPI WebSocket gateway and state machine on Google Cloud Run Gen2 with session affinity, gating inference via MediaPipe Hands so the model evaluates only when the user's hands clear the frame (Python, WebSockets, Docker).	May 2026 – May 2026
Recursive Ray Tracer <i>CSCI 5607, University of Minnesota</i> • Implemented a recursive ray tracer in C++ – ray-primitive intersection, Blinn-Phong shading, reflection/refraction via Schlick's approximation, and Monte Carlo diffuse interreflection.	Jan 2026 – Apr 2026

RESEARCH

Imitation of Imitations: Model Collapse under Synthetic Training Data (in preparation, arXiv) M. Taneja, A. Arya <i>Controlled study of model collapse: pretrained GPT-2 on clean/synthetic code mixtures and evaluated on rare languages (Rust, CUDA, Assembly). Performance degrades gradually up to 90% synthetic data, then collapses sharply (3× perplexity) at 100% — identifying even a small fraction of human data as a critical grounding signal.</i>	2026
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SKILLS

Programming Languages: Python, C++, SQL, JavaScript / TypeScript
ML & Deep Learning: PyTorch, JAX, Hugging Face Transformers, LoRA fine-tuning, scikit-learn, NumPy, Pandas
ML Infrastructure & Serving: Docker, Google Cloud Run, FastAPI, WebSockets, REST APIs, model serving, Git, Linux
NLP & LLMs: LangChain, OpenAI / Anthropic APIs, Agentic systems, synthetic data generation
Computer Vision: OpenCV, torchvision, Ultralytics YOLO, Structure-from-Motion